HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented scalable batch and streaming data pipelines on **AWS** using **Python** and **SQL**, incorporating robust data quality checks and monitoring protocols.
- Developed and deployed interactive **Power BI** dashboards and semantic layers, enabling manufacturing and quality teams to derive actionable insights from production data.
- Engineered and productionized **AI agent** and **LLM** applications, including **RAG** and multi-step workflows, integrated with enterprise data systems to solve manufacturing challenges.
- Collaborated with manufacturing, quality, and engineering teams to frame problems and deliver AI-driven solutions, ensuring alignment with project requirements and business objectives.
- Managed the end-to-end lifecycle of data analytics projects, from initial requirements gathering through to deployment and post-delivery support, adhering to **GMP** data integrity standards.
- Implemented **CI/CD** practices for data pipelines and analytics solutions using **Git** and **Databricks Workflows**, ensuring reliable and repeatable deployments.
- Contributed to the design and implementation of enterprise-scale data solutions on **Azure Databricks**, focusing on performance optimization and maintainability for batch and streaming workloads.

SKILLS

Data Engineering – Azure Databricks, AWS, Python, SQL, Delta Lake, Spark, Batch Processing, Streaming Pipelines, Data Quality, CI/CD, Git

Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Semantic Layers, Self-Service Analytics

AI/ML – AI Agents, LLM, RAG, Multi-step Workflows, Machine Learning Fundamentals, Scikit-learn

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Azure

DevOps & Tools – Databricks Workflows, Airflow, Jenkins, Docker, Git, Jira, Confluence

Databases – PostgreSQL, MySQL, Oracle, SQL

PROJECTS

Manufacturing Data Pipeline and Analytics Platform

- Developed a scalable data pipeline using **Python** and **SQL** on **AWS** to ingest and process manufacturing sensor data, ensuring data integrity and availability for downstream analytics.
- Built a suite of **Power BI** dashboards to visualize key production metrics, enabling quality control teams to identify and address process deviations, reducing defects by approximately 15%.
- Implemented **CI/CD** for pipeline updates using **Jenkins** and **Git**, automating testing and deployment processes to improve delivery speed and reliability.

AI-Powered Anomaly Detection for Production Lines

- Designed and implemented an **AI agent** solution using **Python** and **LangChain** to monitor production line anomalies by integrating with real-time data streams.
- Trained a classification model (scikit-learn) on historical production data to detect anomalies, achieving an **AUC** of 0.88 on a validation set of 50,000 records.
- Deployed the anomaly detection model as a REST API using **FastAPI**, integrated into the existing manufacturing data platform for real-time alerting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)